

Math 515 Qualifying Exam Problems
Exercise 2.6 and Exercise 2.16

Exercise 2.6 1997 Qualifying Exam, Problem 6

For every $\varepsilon > 0$ and every subset E of the real numbers, define

$$\mu_\varepsilon(E) = \inf \sum_{n=1}^{\infty} [\ell(I_n)]^{1/2},$$

where $\ell(\cdot)$ denotes the length of an interval, and the infimum is taken over all countable collections of open intervals $\{I_1, I_2, \dots\}$ which cover E and satisfy

$$\ell(I_n) < \varepsilon \quad \text{for all } n.$$

Prove the following:

- (a) μ_ε is an outer measure on $\mathbb{R}$.
- (b) If $\mu(E) := \sup_{\varepsilon > 0} \mu_\varepsilon(E)$, then $\mu(E) = \lim_{\varepsilon \rightarrow 0^+} \mu_\varepsilon(E)$.
- (c) $\mu((0, 1)) = +\infty$.

(a)

For (a) we need to show $\mu_\varepsilon(\emptyset) = 0$, $E \subseteq F \Rightarrow \mu_\varepsilon(E) \leq \mu_\varepsilon(F)$.

$$\mu_\varepsilon\left(\bigcup_{k=1}^{\infty} E_k\right) \leq \sum_{k=1}^{\infty} \mu_\varepsilon(E_k).$$

$\emptyset$ can be cover by any empty open intervals I_n , whose $\Sigma = 0$
hence $\mu_\varepsilon(\emptyset) = \inf \sum_{i=1}^{\infty} \sqrt{\ell(I_n)} \leq 0$. Since $\mu_\varepsilon(\emptyset) \geq 0$

$$\Rightarrow \mu_\varepsilon(\emptyset) = 0.$$

Monotonicity

For any countable open interval $I = \{I_1, \dots, I_n \dots\}$ of F
is also a countable open cover of E .

Let $C(E)$ and $C(F)$ denote as the collection of all admissible cover for E and F .
Since every admissible cover of F is also admissible cover of E ,

So $C(F) \subseteq C(E)$, Taking inf of a larger collection can only make it smaller, hence

$$\mu_\varepsilon(E) = \inf_{C(E)} \sum_n \sqrt{\ell(I_n)} \leq \inf_{C(F)} \sum_n \sqrt{\ell(I_n)} = \mu_\varepsilon(F)$$

Sub-additive.

Let $E \subseteq \bigcup_{k=1}^{\infty} E_k$, WTS: $\mu_\varepsilon(E) \leq \sum_{k=1}^{\infty} \mu_\varepsilon(E_k)$
RHS diverge to $+\infty$ is trivial.
So assume $\sum_{k=1}^{\infty} \mu_\varepsilon(E_k) < \infty$, then each $\mu_\varepsilon(E_k)$ is finite.

Fix $\delta > 0$, for each k , since $\mu_\varepsilon(E_k)$ is a finite inf,
choose a admissible cover $\{I_{kn}\}_{n=1}^\infty$ of E_k s.t

$$\sum_{n=1}^\infty \sqrt{\ell(I_{k,n})} \leq \mu_\varepsilon(E_k) + \frac{\delta}{2^k}.$$

Combining all these interval gives the countable family $\{I_{kn}\}_{k,n=1}^\infty$
It covers $\bigcup_k E_k \supseteq E$, and every interval still has length $< \varepsilon$, so it is
an admissible cover of E . Since $\mu_\varepsilon(E)$ is the inf over all admissible covers

$$\mu_\varepsilon(E) \leq \sum_{k=1}^\infty \sum_{n=1}^\infty \sqrt{\ell(I_{k,n})} \leq \sum_{k=1}^\infty \left(\mu_\varepsilon(E_k) + \frac{\delta}{2^k} \right) = \sum_{k=1}^\infty \mu_\varepsilon(E_k) + \delta$$

Since $\delta > 0$ is arbitrary, letting $\delta \rightarrow 0^+$ yield

$$\mu_\varepsilon(E) \leq \sum_{k=1}^\infty \mu_\varepsilon(E_k)$$

(b)

For any $0 < \delta < \varepsilon$, then any countable open covering $\{I_1, \dots, I_n, \dots\}$ with $\ell(I_n) < \delta$, It is a also a
countable open covering $\{I_1, \dots, I_n, \dots\}$ with $\ell(I_n) < \varepsilon$.

$$\begin{aligned} \text{So } \mu_\delta(E) &= \inf_{\ell(I_n) < \delta} \sum_{n=1}^\infty \sqrt{\ell(I_n)} \geq \inf_{\ell(I_n) < \varepsilon} \sum_{n=1}^\infty \sqrt{\ell(I_n)} = \mu_\varepsilon(E) \\ &\Rightarrow \mu_\delta(E) \geq \mu_\varepsilon(E) \end{aligned}$$

By Monotonicity of $\mu_\varepsilon(E)$ in ε , the one-sided limit exists and equals the sup.

$$\Rightarrow \lim_{\varepsilon \rightarrow 0^+} \mu_\varepsilon(E) = \sup_{\varepsilon > 0} \mu_\varepsilon(E).$$

(c)

For each covering $\{I_n\}$

$$\begin{aligned} &\ell(I_n) < \varepsilon \\ &\Rightarrow \sqrt{\ell(I_n)} < \sqrt{\varepsilon} \\ &\Rightarrow \frac{1}{\sqrt{\ell(I_n)}\sqrt{\varepsilon}} < \frac{1}{\ell(I_n)} \quad \text{divide both side by } \ell(I_n) \cdot \sqrt{\varepsilon}. \\ &\Rightarrow \sqrt{\ell(I_n)} \cdot \sqrt{\varepsilon} < \ell(I_n) \\ &\Rightarrow \sqrt{\ell(I_n)} > \frac{\ell(I_n)}{\sqrt{\varepsilon}} \\ &\sum_{n=1}^\infty \sqrt{\ell(I_n)} > \sum_{n=1}^\infty \frac{\ell(I_n)}{\sqrt{\varepsilon}}; \quad \text{Since } \{I_n\} \text{ over } (0, 1) \\ &\text{So } \sum_{n=1}^\infty \ell(I_n) \geq 1, \text{ hence } \mu_\varepsilon((0, 1)) \geq \frac{1}{\sqrt{\varepsilon}}. \end{aligned}$$

Note $\varepsilon \rightarrow 0^+$, $\frac{1}{\sqrt{\varepsilon}} \rightarrow \infty$.

So $\mu_\varepsilon((0, 1)) \rightarrow \infty$, Thus $\mu((0, 1)) = \infty$.

Exercise 2.16 2000 Qual Problem 1, rephrased

Define the measure λ on the Borel σ -algebra of $\mathbb{R}$ by

$$\lambda(A) = \mu(A \cap (0, 1)),$$

where μ is Lebesgue measure. Let

$$K = \bigcap \{A : A \text{ is closed and } \lambda(A) = 1\},$$

and

$$D = \bigcap \{G : G \text{ is open and } \lambda(G) = 1\}.$$

Determine precisely which points belong to K and D .

K :

For any A with $\lambda(A) = 1$.

$$\mu(A \cap (0, 1)) = 1, \text{ Since } \mu((0, 1)) = \mu(A \cap (0, 1)) + \mu((0, 1) \setminus A) \Rightarrow 1 = 1 + \mu((0, 1) \setminus A)$$

So $\mu((0, 1) \setminus A) = 0$.

So for any $x \in (0, 1)$

If $x \notin A$, Since A is closed, then $(0, 1) \setminus A$ is open.

$x \in (0, 1) \setminus A$. So there exist $\delta > 0$, s.t

$$(x - \delta, x + \delta) \subseteq (0, 1) \setminus A.$$

But $\mu((x - \delta, x + \delta)) = 2\delta > 0$.

$$\mu((0, 1) \setminus A) \geq \mu(x - \delta, x + \delta) > 0.$$

$$\Rightarrow \mu((0, 1) \setminus A) = 0. \text{ So } x \in A. \text{ Note } A \text{ is closed, So } [0, 1] \subseteq A.$$

Hence $[0, 1] \subseteq K$, Since $\lambda([0, 1]) = 1$, So $[0, 1] = K$.

D :

Taking any $x \in (0, 1)$.

Let $G = (0, 1) \setminus \{x\}$. Since $\{x\}$ is closed hence G is open.

Since $\mu(\{x\}) = 0$, So $\mu((0, 1) \setminus \{x\}) = \mu(0, 1) \setminus \mu(\{x\}) = 1 - 0 = 1$. (i.e. $\lambda(G) = 1$.)

Note $x \notin G$, $x \notin \bigcap \{G : G \text{ is open, } \lambda(G) = 1\} = D$

No point of $(0, 1)$ in D , and $\lambda((0, 1)) = 1$. Force $D = \emptyset$